Schmucker, U.: Erdmagnetische Tiefensondierung in Deutschland 1957/59. Magnetogramme und erste Auswertung. — Abh. Akad. Wiss. Göttingen, Math.-Phys. Kl., Beitr. zum Internationalen Geophysikalischen Jahr, 5, 1959.

—: Anomalies of Geomagnetic Variations in the Southwestern United States. — J. Geomagnetism and Electricity, **XV**, 4, 193—221, 1964.

Wiese, H.: Geomagnetische Tiefentellurik. — Deutsch. Akad. Wiss. Berlin, Geomagn. Inst., **36**, Akademie-Verlag, Berlin (zusammenfassende Darstellung mit ausführlichem Literaturverzeichnis), 1965.

Palaeomagnetic investigations of the tertiary and quaternary igneous rocks: III A palaeomagnetic study of the East Slovak Province

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With 5 figures and 3 tables

Zusammenfassung

Innerhalb der Grenzen, die durch stratigraphische Probleme hervorgerufen werden, wird der Versuch gemacht, eine paläomagnetische Abfolge des Sarmats zu geben und als Arbeitshypothese die Korrelation mit Japan und Neu-Zeeland vorgeschlagen. Die Homogenität der Intensität und Susceptibilität der Vulkanite der östlichen Slowakei läßt vermuten, daß das Muttermagma wenig Änderungen in der Periode der Aktivität trotz der petrographischen Unterschiede der Vulkanite besaß.

Abstract

Within the limits imposed by stratigraphical uncertainties, a tentative palaeomagnetic sequence is suggested for the Sarmatian. Correlation with New Zealand and Japan is proposed as a working hypothesis. The homogeneity of the intensity and susceptibility of magnetization of the volcanics of East Slovakia suggests that the parent magma may have suffered little contamination during the period of activity despite the petrological variety.

Résumé

Entre les limites imposées par les incertitudes stratigraphiques une série tentative de zones paléomagnétiques est proposée. Comme un premier essai, une correlation avec la Nouvelle Zélande et le Japon est supposée. La répartition homogène de l'intensité et de la susceptibilité des volcaniques de la Slovakie de l'est conduit a l'idée que le magma parent n'a subi que des variations infimes pendant la période de l'activité malgré la diversité types pétrographiques.

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Краткое содержание

Автор приводит результаты палеомагнитных исследований третичных и четвертичных вулканитов восточной Чехословакии и сравнивает их с вулканитами Японии и Новой Зеландии.

Introduction

The volcanic rocks of the East Slovakian province lie some 250 km. east of Banská Štiavnica, the centre of the Central Slovakian province, and at their nearest point are only some 20 km. from the Russian frontier. In general terms they form the low range of hills extending from the region of Prešov in the north to the Hungarian border at a point south-east of Košice. Isolated outliers are found in the basin to the east of the hills near to, and south of, Michalovce. The line of hills is broken by a single, narrow, molasse filled depression almost due east of Košice separating the hills to the north, the Slaské Pohorie, from those to the south known as the Velký Milič, whose continuation into northern Hungary terminates southwards in Tokaj.

The whole upland region is densely wooded and traversed by few roads. Although the area is still little developed, roads encircle the hills and traverse the molasse filled depression, and these give access to peripheral quarries which offer about the only possibilities for sampling. The distribution of these quarries is clearly shown in fig. 1. In this densely wooded area natural outcrops are rare, difficult to find, and usually remote. The samples on which this account is based were collected during the summer of 1963 and in the Spring of the following year. The programme was arranged with the help of Dr. M. Kuthan director of the Geologický ústav Dionýza Štura Bratislava, whilst Dr. K. Karolus provided invaluable help in the field and also gave the necessary geological information. Their help and that of RN Dr. J. Mašin, Ing. M. Krs and S. Duratny and the general interest and cooperation of their colleagues is gratefully acknowledged.

Outline of geology

The Slaské Pohorie and Velký Milič and their Hungarian extension, the Zemplenihegyseg, lie within the arc of the Carpathian Mountain System of which they form part. To the east lies the Michalovce Basin, in which volcanic outliers are found, bounded on its eastern side by the Vihorlat Hills which pass into Russia near Užgorod. To the west the basin in which Košice and Prešov lie, is bounded by the Spišso-gemerské rudéhorie. Both basins have a fill of late Tertiary molasse.

As in Central Slovakia, so too in the volcanics of Eastern Slovakia, a succession of petrographical types can be recognised. The general sequence of phases is similar although there appears to be no third rhyolite phase in East Slovakia and no introductory andesite phase in Central Slovakia. The similarity in the succession of phases does not imply a time equivalence and indeed one of the major objects of research is to establish whether either is a time sequence in the absence of a through section and with good stratigraphical marker horizons rare. It is however con-

Table 1. Age and possible correlation of the Central and East Slovakian andesites.

		Central Slovakia		East Slovakia proposed correlation (KUTHAN 1948, 1964)	East Slovakia former correlation
Sarmatian	Upper		R	2nd. phase andesites	
	Middle	Final phase andesites Third phase rhyolites	N R	1st. phase andesites	
	Lower	3rd. phase andesites intrusives 2nd. phase andesites	N	2nd. phase rhyolites Introductory phase andesites	2nd. phase andesites 1st. phase andesites
Tortonian	Upper		R		
	Middle	2nd. phase rhyolites			2nd. phase rhyolites
	Lower	1st. phase andesites	R		Introductory phase andesites
Helvetian					
Burdigalian		1st. phase rhyolites		1st. phase rhyolites	1st. phase rhyolites

venient to know that similarly numbered phases are petrographically similar, for example the first phases andesites characteristically are garnetiferous.

One consequence of the stratigraphical problem is that dating and correlation are necessarily tentative, and table 1 illustrates two correlations

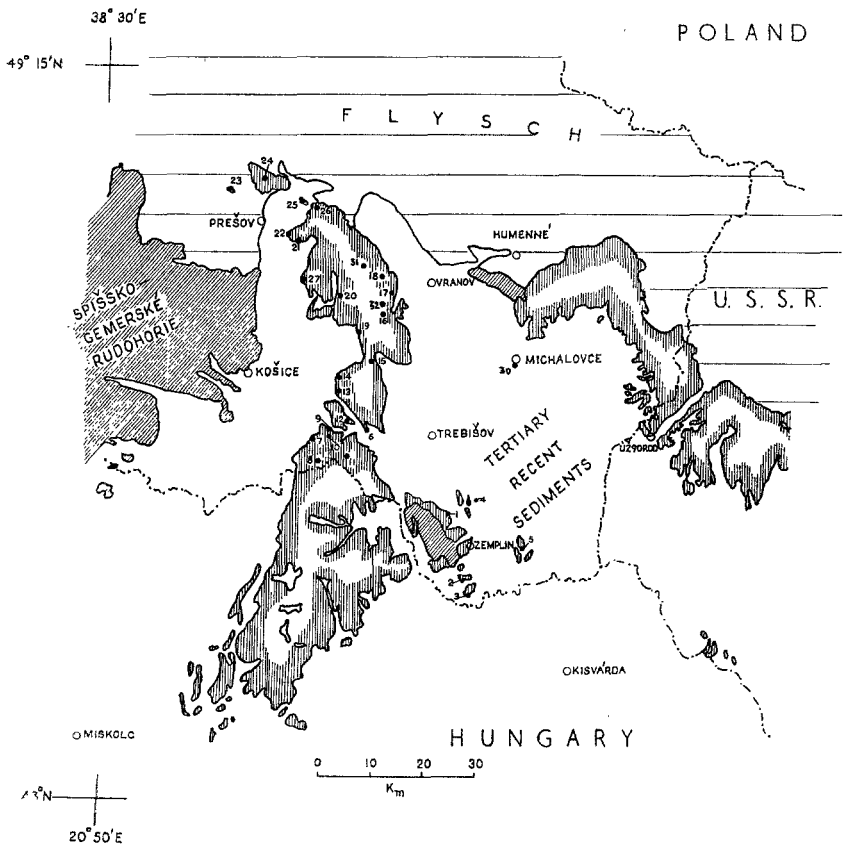


Fig. 1. Simplified outline map of East Slovakia indicating the location of sites. Outcrops of andesites indicated by vertical ruling.

which have been proposed. The correlation here adopted is due to the current research of Dr. KUTHAN and his colleagues (see KUTHAN 1948, 1964). The first rhyolite phase is found associated with dated Burdigalian conglomerates and would therefore seem coeval with the same phase in Central Slovakia. The second rhyolite phase is not so well dated for the associated micro-fauna may range in age from upper Helvetian to lower Sarmatian. It was formerly correlated with the second rhyolite phase in Central Slovakia known to be middle Tortonian in age from evidence in

Table 2. Site Mean Directions of Magnetization and statistics of rocks from the

1 Site no.	2* Site and/or rock type	3 Magnetization Declination	Inclination	4 Sampling a/b	5 Confidence α
1	Kašov: 2nd. phase rhyolite	279.1	+ 31.1	6 — 6	10.2
2	Středa nad Bodrogom: 2nd. ph. andesite	140.9	— 53.5	5 — 6	15.9
3	Malý Kamenec: 2nd. ph. rhyolite	333.6	+ 60.3	6 — 6	4.4
4	Sirník: 2nd. ph. andesite	236.8	— 57.4	2 — 6	8.4
5	Svatuše (Plešany): 2nd. ph. andesite	167.8	— 23.6	6 — 6	9.9
6	Kalša: introductory ph. andesite	333.9	+ 59.3	5 — 6	result
7	Nižna Myšla: introd. ph. andesite	307.6	+ 72.9	5 — 6	3.2
8	Skaroš: introd. ph. andesite	200.8	— 75.3	6 — 6	5.4
9	Rakoš: introd. ph. andesite	345.4	+ 65.1	6 — 6	4.4
10	Vyšná Myšla: dacite between introd. and 1st. ph. andesite	27.9	+ 67.3	3 — 6	9.6
11	Malá Milič: dacitoid andesite type Milic	218.0	— 74.5	3 — 6	17.6
12	Slanec: introd. ph. andesite	352.8	+ 63.0	3 — 6	result
13	Ruskov: 2nd. ph. andesite	39.7	+ 69.9	5 — 6	6.7
14	Durkov: 2nd. ph. andesite	31.5	+ 42.2	5 — 6	8.6
15	Drahov: 2nd. ph. andesite	350.2	+ 57.9	4 — 6	11.3
16	Banské: 2nd. ph. andesite	352.8	+ 20.3	5 — 6	8.7
17	Vechec-Kamenná: 2nd. ph. andesite	185.3	— 52.4	6 — 6	3.9
18	Juskova Volá: latite 2nd. rhyolite ph.	202.0	— 62.8	5 — 6	8.1
19	Vizná Kamenica: 2nd. ph. andesite	213.9	— 53.7	2 — 6	22.4
20	Opiná: 2nd. ph. andesite	188.7	— 59.2	6 — 6	2.1
21	Kokošovce: 2nd. ph. andesite	289.0	+ 58.7	5 — 6	19.1
22	Ruská Nová Ves: 2nd. ph. andesite	354.5	+ 66.4	6 — 6	7.2
23	Velký Šariš: 1st. ph. andesite	174.6	— 73.3	6 — 6	6.7
24	Fintice: 1st. ph. andesite	172.0	— 69.1	6 — 6	5.0
25	Vyšná Šebastova: 1st. ph. andesite	351.2	+ 68.1	6 — 6	7.8
26	Borovnyk: 1st. or 2nd. ph. andesite	42.3	+ 70.1	6 — 6	4.7
27	Brestov: 1st. ph. andesite	103.2	— 70.3	5 — 6	9.3
28	Žehna: 2nd. ph. andesite	3.0	+ 67.7	6 — 6	14.0
29	Červenica: 2nd. ph. andesite	205.6	— 65.7	5 — 6	11.2
30	Michalovce: 2nd. ph. rhyolite	190.2	— 47.9	5 — 7	3.3
31	Zámutov: 2nd. ph. andesite	54.1	+ 52.8	7 — 7	11.7
32	Banské: 2nd. ph. andesite	188.0	— 58.6	6 — 6	10.3
33	Hermanovce: 2nd. ph. andesite	357.1	+ 59.7	7 — 7	5.6

Column 2 indicates the site and the petrological description of the rock type; Column + north seeking pole pointing downwards — when pointing upwards. Column 4 shows collected (b). Column 5 gives the semi angle of the cone of confidence α calculated from error. Column 7 R is the length of the vector mean computed in the calculation of α. Column 9 gives the long mean and standard deviation of sample intensities of natural mean susceptibility and its standard deviation.

Hungary. Current opinion would place the rhyolites in the lower Sarmatian at the top of their possible time range thus regarding the introductory phase andesites as spanning the Tortonian—Sarmatian boundary. As a consequence of this the second phase andesites are placed in the middle to upper Sarmatian and thus younger than the final phase andesites of Central Slovakia.

East Slovakian Province

6 C.S.D.	C.S.E.	7 Vector R	8 Demagnetizing Field gauss.	9 Log. mean Intensity and st. deviat. $\times 10^{-4}$ gauss. N. R. M.		Min. scatter		10 Log. mean Suscept. $\times 10^{-4}$ oe/gauss. and st. deviat.	
12.3	5.0	5.886	170	not measured		0.42	0.40	not measured	
16.5	7.4	4.834	85	not measured		1.90	0.39	not measured	
5.3	2.2	5.979	255	1.12	0.15	0.70	0.26	0.58	0.09
7.7	6.2	1.988	340	0.12	0.07	0.05	0.17	0.34	0.08
11.9	4.9	5.892	85	0.33	0.11	0.48	0.09	0.97	0.09
not significant			170	0.63	0.07	0.57	0.11	0.58	0.11
3.4	1.5	4.993	85	0.26	0.77	$\bar{1}$.01	0.72	0.04	0.38
6.6	2.7	5.967	85	0.65	0.28	0.67	0.31	0.89	0.06
5.3	2.2	5.979	255	0.47	0.22	0.40	0.24	$\bar{1}$.98	0.28
6.3	3.6	2.988	340	0.85	0.19	0.84	0.37	0.91	0.27
11.5	6.6	2.960	85	0.07	0.14	0.04	0.16	0.77	0.07
not significant			170	0.47	0.09	0.21	0.07	0.95	0.15
7.1	3.2	4.969	255	not measured		not measured		0.06	0.22
9.1	4.1	4.950	0	0.69	0.08	0.69	0.08	0.72	0.08
9.9	5.0	3.955	340	0.25	0.29	0.15	0.29	$\bar{1}$.82	0.09
9.2	4.1	4.949	170	$\bar{1}$.94	0.06	$\bar{1}$.71	0.13	$\bar{1}$.76	0.19
4.8	1.9	5.983	340	0.74	0.35	0.43	0.34	$\bar{1}$.49	0.07
8.5	3.8	4.956	340	not measured		not measured		$\bar{1}$.90	0.40
7.2	5.1	1.992	0	not measured		not measured		0.05	0.02
2.5	1.0	5.995	0	0.69	0.41	0.69	0.41	1.32	0.04
19.8	8.8	4.764	0	0.53	0.15	0.53	0.15	0.36	0.04
8.6	3.5	5.943	85	0.51	0.12	0.50	0.16	$\bar{1}$.38	0.18
8.0	3.3	5.951	0	$\bar{1}$.87	0.75	$\bar{1}$.87	0.75	0.64	0.04
6.0	2.5	5.973	0	0.58	0.06	0.58	0.06	0.59	0.04
9.4	3.8	5.933	85	not measured		not measured		0.03	0.18
5.6	2.3	5.976	85	1.73	0.13	1.67	0.11	0.86	0.02
9.8	4.4	4.942	85	0.18	0.09	0.34	0.06	1.06	0.03
16.6	6.8	5.790	85	0.70	0.14	0.40	0.14	1.28	0.06
11.7	5.3	4.916	85	0.89	0.05	0.73	0.09	$\bar{2}$.60	0.14
3.5	1.6	4.992	170	not measured		not measured		0.73	0.31
15.5	5.8	6.782	170	0.65	0.40	0.41	0.29	$\bar{1}$.14	0.27
12.4	5.0	5 884	0	1.87	0.39	1.87	0.39	1.03	0.42
7.5	2.8	6.949	255	0.96	0.19	0.63	0.24	not measured	

3 the mean direction of magnetization the sign of inclination following the convention of the number of samples used in the computation (a) and the total number of samples FISHER. Column 6 C.S.D. = circular standard deviation, C.S.E. = circular standard Column 8 gives the magnitude of the demagnetizing field at which C.S.D. is a minimum. remanent magnetization and when C.S.D. is a minimum. Column 10 indicates the log

Palaeomagnetic investigation

a) Direction of magnetization. One or two cores were prepared from each of the six or seven orientated blocks collected from each of the thirty three field sites. Although the number of sites is small, it is a direct reflection of the problems of sampling in East Slovakia. For the most part the direction of magnetization of a single cylinder from each

sample was determined under a short period low sensitivity astatic magnetometer. Samples were then progressively demagnetized in alternating magnetic fields of up to 350 gauss in four stages, the direction and intensity of magnetization being measured after each step. Site mean directions of magnetization were computed for each stage using a computer pro-

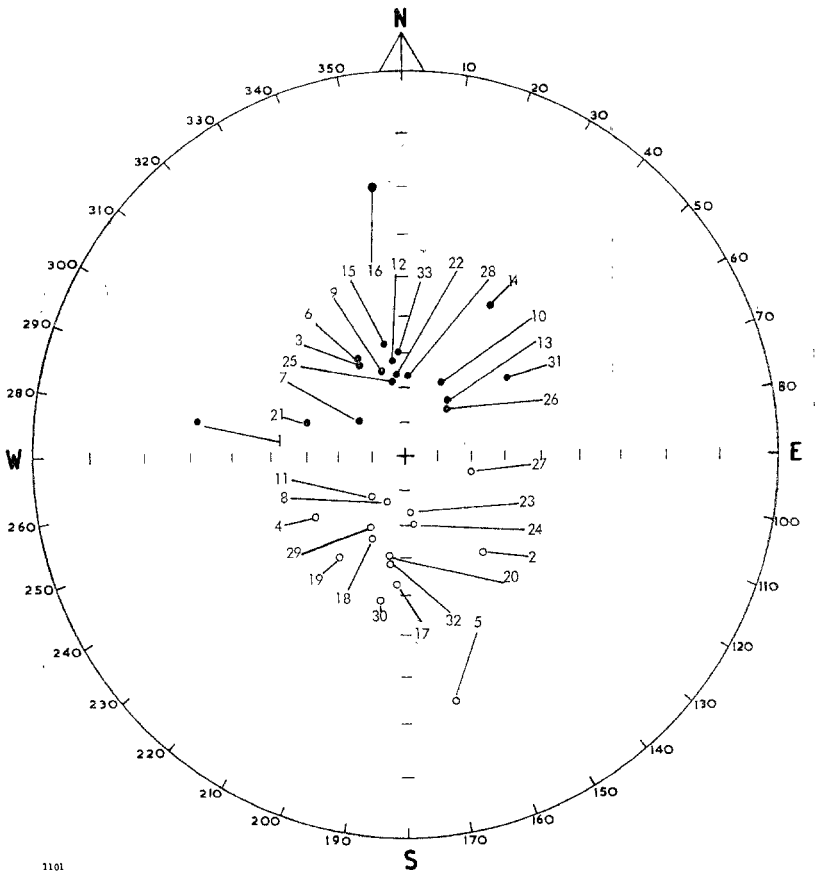


Fig. 2. Stereogram of the site mean directions of magnetization of the East Slovakian Volcanics. Site numbers corresspond to the numbers in table 1.

gramme slightly different from that used on the Central Slovakian data in that minimum value of the circular standard deviation was used as the best estimate of the original direction of magnetization instead of the k value. When this direction of least scatter coincided with the measurement of natural remanent magnetization no significant change in direction occured during the demagnetization process.

In some cases samples were too weakly magnetized for measurement or gave inconsistent results (in such cases the intensity of magnetization was

usually near the lower limit of measurement) under the low sensitivity magnetometer. Such samples were then measured under a high sensitivity instrument after each of the first two demagnetization stages. The site mean directions and relevent statistics are given in table 2 and the directions illustrated in fig. 2.

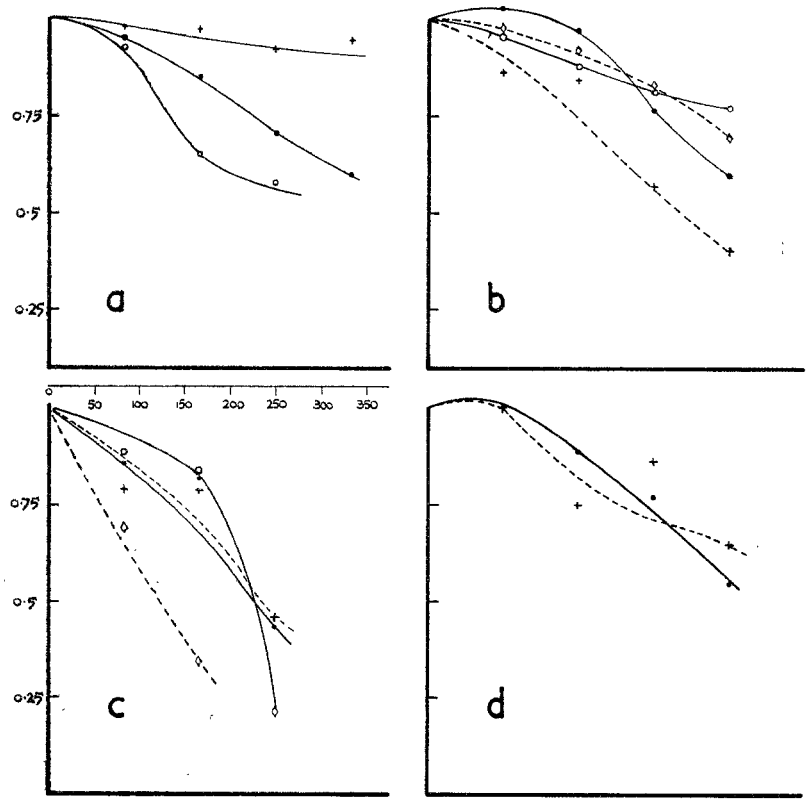


Fig. 3. Normalized alternating current demagnetization curves of East Slovakian volcanic rocks. — a) Sites in introductory phase volcanics. b) ... second phase andesites. c) ... first phase andesites. d) ... second phase rhyolites.

b) Intensity of magnetization. The intensity of magnetization was measured concurrently with the direction of magnetization. Normalized demagnetization curves for a number of sites, normal and reversed, from various petrological phases are shown in fig. 3. The mean intensity value and its standard deviation for the natural remanent magnetization for each site are shown in table 2 in log. form; the value of intensity at the stage of minimum scatter is also given. An attempt to show the distribution of these intensities through the igneous suite as a whole is shown in fig. 4, in the form of semi-log. cumulative percentage curves. The curves are based on individual samples rather than site means.

c) Susceptibility of magnetization. The susceptibility of magnetization of the samples in low fields (of the order of 2 gauss) was measured on a transformer bridge apparatus (COLLINSON, MOLYNEUX, STONE 1963) and the site mean and standard deviation calculated. These figures, also in log. form are given in table 2, and their distribution through the igneous suite is illustrated in fig. 5, a cumulative percentage curve as fig. 4.

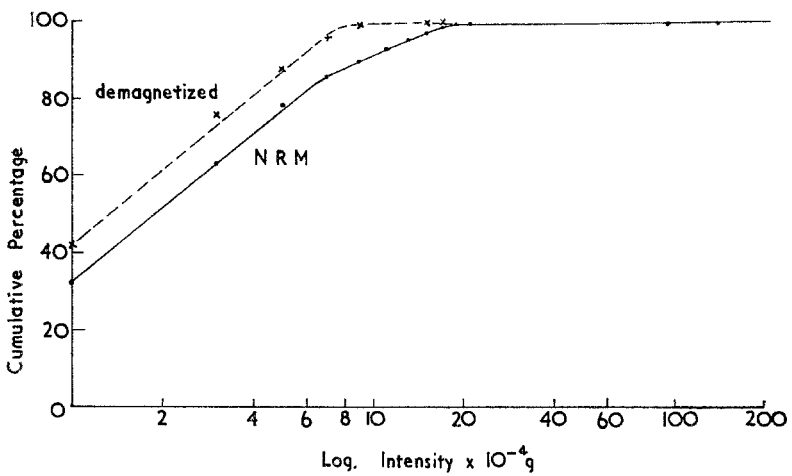


Fig. 4. Cumulative percentage curves showing the distribution of intensity of magnetization in the rocks of the East Slovakian igneous suite. Curves are plotted on semi-log. paper.

Discussion of results

a) Directions of magnetization. The scatter about the site mean directions is in general considerably greater than that commonly found in more basic rocks, although the soft, secondary component of magnetization is comparatively small. Arbitrarily values of circular standard deviation greater than 20 have been regarded as not significant. This affects two sites (6 and 12) and these are so indicated in table two although the mean directions are similar to the remaining sites (see fig. 2). At three sites low inclinations have been recorded with values of from 20° to 30°.

The difficulty accompanying any attempt to set up a tentative sequence of zones of normal and reversed magnetization quite apart from the stratigraphical problems, is the lack of evidence of relative age differences between sites in rocks belonging to the same petrographic phase. The scheme proposed for the Sarmatian under the new correlation of the volcanics with Central Slovakia, depends upon the validity of that correlation for it makes use of the somewhat better, if still tentative, arguments in favour of the zonal sequence established there. The critical point is that what are suggested as the younger second phase andesites, the older third phase andesites, and the intrusive phase regarded as belonging to the time interval between

the two phases are all normally magnetized (NAIRN 1966 in press). Thus the correlation of the second phase rhyolites of East Slovakia with the third phase andesites automatically fixes the stratigraphical position of the first zone of normal magnetization. From this the rest follows, and the simplest sequence of zones during the Sarmatian consistent with the observations, consists of two normal and two reversed zones.

The attempt to equate this tentative alternation of zones of normal and reversed magnetization in the Sarmatian with sequences in other regions is complicated in the first place by stratigraphical uncertainties. It seems

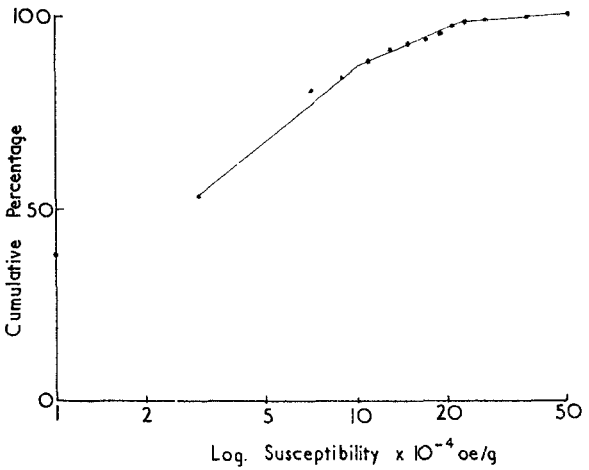


Fig. 5. Cumulative percentage curve showing the distribution of susceptibility in rocks of the East Slovakian igneous suite. Curve is plotted on semi-log. paper.

(GLAESSNER 1953) that GIGNOUX (1950) was in error in regarding the Sarmatian as in part a Tortonian facies equivalent, and that in the region of the Vienna basin the Sarmatian should certainly be regarded as post Tortonian. The succeeding epoch, the Pannonian, defined in the Hungarian Basin, FUNNELL (1964) equates to Pontian s. l. and includes both it and the Sarmatian in the Upper Miocene, whilst GIGNOUX considers the Pontian as equal to the whole of the Upper Miocene time interval. FUNNELL however adopts an age of 7 my. as the base of the Pliocene and 12 my, as the base of the upper Miocene whilst HOLMES (1959) places the base of the Pliocene at 11 my., EVERNDEN et al. (1961) at 12 my., and KULP (1961) at 13 ± 1 my., respectively.

In the circumstances any attempts at palaeomagnetic correlation have only value as discusssion models, anything further without a detailed study of the whole Miocene—Pliocene boundary problem would be premature. If then the Sarmatian is equated empirically with the Upper Miocene which seems to conform to Eastern European usage, the palaeomagnetic zonal sequence may be compared with results from other areas. As will be seen in table 3, only the New Zealand and Japanese results can be

directly compared; elsewhere either the stratigraphy or the palaeomagnetic data is not given in a form suitable to attempt correlation. Thus in Iceland, in the Merkjargil section HOSPERS (1953) records the same zonal sequence but gives the age merely as „probably Miocene", and in the western U.S.A., WATKINS (1965) reports five pairs of normal and reversed magnetization in one section with radiometric dates of 15.4 and 14.5 my. for one normal zone without it being too clear whether this is near the top or the bottom of his series. In France ROCHE (1960) records both normal and reversed magnetization, although in an earlier work only one normal zone was reported (see RUTTEN 1960).

Table 3. Possible palaeomagnetic correlation of the top part of the Miocene.

	Slovakia	New Zealand	Japan
Upper	R		Itaorishi fm.
Sarmatian Middle	N	Third eruptive ph	Shirasaka fm.
	R	Second eruptive ph	N
Lower	N	First eruptive ph	Omame-zawa fm.
Tortonian Upper	R		Yumoto fm.

In the Dunedin Volcanics, COOMBES and HATHERTON (1959) report two normal and a reversed zone in rocks considered as late middle to upper Miocene in age. These then correlate with the lower part the Sarmatian. The Japanese result (NOMURA 1963) in which only two zones are recorded from the three formations forming the Upper Miocene presents a problem. It suggests that either a zone of reversed magnetization has been missed between the Omame-zawa and the Shirasaska formations (see table 3), or that the first two zones in Slovakia and New Zealand be relegated to the top Tortonian, or that in the latter areas self reversal may have occurred.

b) Intensity and susceptibility of magnetization. Compared with susceptibility, there is as in Central Slovakia, a much greater scatter in the intensity values about the site mean (see table 2). This seems to suggest that the difference in intensity between samples at a given site is not so much related to variations in the ferromagnetic mineral content as to their state of magnetization. Although no rigid pattern of sampling was followed, collecting was in general spread over the available outcrop, so that the intensity variability may indicate the high degree of sensitivity of ferromagnetic minerals to even very local difference in the thermal or mineral oxidation history, even if it is hard to see how these would arise. It is somewhat different from the changes in intensity noted with distance from the flow surface (ANGENHEISTER and TURKOWSKY 1964).

It is interesting to note that the susceptibility distribution curve fig. 5, shows one inflection point which does not appear in the demagnetized intensity curve fig. 4. There appears to be about 2% with very high susceptibility and intensity values. Both sets of curves are reasonable approximations to a log normal distribution. The homogeneity of the intensity and susceptibility suggests that despite the variety of petrographical types recognised the whole sequence forms a unity, with possibly only a small composition change as indicated by the susceptibility distribution. The amount of material available precludes the investigation of distributions of these quantities within individual petrological phases.

Bibliography

ANGENHEISTER, G. & TURKOWSKY, C.: Die Verteilung der induzierten und natürlichen Magnetisierung innerhalb einiger Basaltlagen des Vogelsberges. — Boll. Geofisica Teor. ed Applicata, **6**, 285—295, 1964.

COLLINSON, D. W., MOLYNEUX, L. & STONE, D. B.: A total and anisotropic magnetic susceptibility meter. — Journ. Sci. Inst., **40**, 310—312, 1963.

COOMBES, B. S. & HATHERTON, T.: Paleomagnetic studies in Cenozoic volcanic rocks in New Zealand. — Nature, **184**, 883—884, 1959.

EVERNDEN, J. F., CURTIS, G. H., OBRADOVICH, J. & KISTLER, R. W.: An evaluation of glauconite and illite for dating sedimentary rocks by the potassium-argon method. — Geochim. et Cosmochim. Acta, **23**, 78—99, 1961.

FUNNELL, B. M.: The Teriary Period, in The Phanerozoic Time Scale. — Geol. Soc. London, 179—191, 1964.

GIGNOUX, M.: Geologie Stratigraphique. — Masson, Paris 1950.

GLAESSNER, M. F.: Time Stratigraphy and the Miocene Epoch. — Bull. Geol. Soc. America, **64**, 647—658, 1953.

HOLMES, A.: A revised geological time scale. — Trans. Edin. Geol. Soc., **17**, 183—216, 1959.

HOSPERS, J.: Reversals of the main geomagnetic field I & II. — Koninckl. Ned. Akad. Wet., B, **56**, 467—491, 1954.

KULP, J. L.: Geologic time scale. — Science, **133**, 1105—1114, 1961.

KUTHAN, M.: Undulačný vulkanizmus karpatského orogénu a vulkanologice štúdie v servernej časti Prešovských hôr. — Práce ŠGÚ, zošit 17, Bratislava, 1948.

—: Vysvetlivky y prehladnej geologickej mape ČSSR 1 : 200,000 list Zborov. — Košice, 35—38, Bratislava, 1964.

NAIRN, A. E. M.: Palaeomagnetic Investigations of the Tertiary and Quaternary Igneous Rocks: II A palaeomagnetic study of the Central Slovakian Province. — Geofys. Sbor. Č. S. A. V., 1966 in press.

NOMURA, S.: Paleomagnetic studies on the Neogene Volcanic Rocks of the Sendai District, Northeast Honshu, Japan. — Earth Science, Journ. Assoc. Collab. Japan, no. 67, 30—39, 1963.

ROCHE, E.: Sur l'aimantation de laves Miocènes d'Auvergne. C. R. Acad. Sci. Paris, **250**, 377—379, 1960.

RUTTEN, M. G.: Paleomagnetic dating of younger volcanic series. — Geol. Rundsch., **49**, 161—167, 1960.

WATKINS, N. D.: Paleomagnetism of the Columbia Plateaux. — Journ. Geophys. Res., **70**, 1379—1406, 1965.